

Practical No. 03

Switching, VLANs, and Inter VLAN Routing on Cisco Packet Tracer

Easiest and complete revision notes, written for someone starting with zero networking knowledge

What This Practical Is About, in One Paragraph

You take four PCs and one switch and split them into two separate logical groups called VLANs, even though all four PCs are plugged into the very same switch. PC1 and PC2 belong to one group, PC3 and PC4 belong to another group. Devices inside the same VLAN can talk to each other immediately. Devices in different VLANs cannot talk to each other at all, until a router is added and configured to allow traffic to pass between the two VLANs. That last part, using one router to connect multiple VLANs through a single cable, is called Router on a Stick, and it is the whole point of this lab.

Key Terms Explained First, Before Anything Else

VLAN, Virtual LAN. A way of grouping devices into their own separate logical network using software settings on a switch, even though the devices are physically plugged into the same switch. Each VLAN behaves like its own independent small network.

Broadcast domain. A group of devices that all receive each other's broadcast messages. Every VLAN is its own separate broadcast domain, meaning a broadcast sent by a device in VLAN 10 is never seen by a device in VLAN 20.

Subnet. A smaller, defined range of IP addresses. In this lab, VLAN 10 uses the 192.168.1.x range and VLAN 20 uses the 192.168.2.x range, so each VLAN is also its own separate subnet.

Access port. A switch port connected to a normal end device, like a PC. An access port belongs to exactly one VLAN only, and simply hands that device's traffic to that VLAN.

Trunk port. A switch port that is connected to another switch or to a router, and is able to carry traffic belonging to many different VLANs over one single cable at the same time, by tagging each frame with its VLAN number.

Trunking / dot1q tagging. The process of stamping a small tag on every frame that passes through a trunk port, stating which VLAN that frame belongs to, so the device on the other end knows how to sort it. dot1q, sometimes written 802.1Q, is simply the standard naming for this tagging method, this is why the router command later says encapsulation dot1q.

Subinterface. A logical, software created division of one single physical interface on a router. Instead of buying two separate physical network cards for the router, you split one physical port, Fa0/0, into two virtual ones, Fa0/0.10 and Fa0/0.20, and each one is dedicated to one VLAN.

Router on a Stick, ROAS. A setup where a single physical cable connects a router to a switch trunk port, and the router uses subinterfaces to separately handle each VLAN's traffic over that one shared cable. This is exactly the setup built in this lab.

Default gateway. The IP address a device sends its traffic to whenever the destination is outside its own subnet. For any PC in VLAN 10 the default gateway is the router subinterface handling VLAN 10, and for VLAN 20 it is the router subinterface handling VLAN 20.

CLI, Command Line Interface. A text based screen where you type commands directly to configure a device, instead of clicking buttons. Both the switch and the router are configured through their own CLI in this lab. Where to find it is explained in the very next section, since this is exactly where most confusion happens.

Running config. The configuration currently active in a device's memory right now. If the device loses power without saving, this configuration is lost completely.

Startup config. The configuration saved permanently to the device's storage, which is loaded automatically every time the device restarts. Saving your work means copying the running config into the startup config.

Ping. A simple test command that sends a small message to another device and waits for a reply, used here to check whether two devices can actually reach each other after all the configuration is done.

Where Exactly Is the CLI, Step by Step

This is the part that usually causes the most confusion, so here it is very slowly, for both the switch and the router.

1. On the Cisco Packet Tracer workspace, left click once on the switch icon (or the router icon).

2. A small window pops up. At the top of that window you will see a few tabs, usually named Physical, Config, and CLI.
3. Click on the tab named **CLI**.
4. A black terminal style screen appears, this is the actual CLI. This is where every command below is typed.
5. If the screen shows a message asking about initial configuration dialog, just type **no** and press Enter, then press Enter again to get a clean prompt.

Very important habit, type each command yourself, one line at a time, and press Enter after every single line. Do not paste a big block of many lines together, Packet Tracer's CLI can misread pasted blocks, and more importantly, typing line by line is how you actually learn what each command does instead of just copying blindly.

The Topology You Are Building

One switch in the middle, four PCs connected to it, and one router connected to the switch through a single trunk cable.

- PC1 and PC2 connect to switch ports Fa0/1 and Fa0/2, these two ports belong to VLAN 10.
- PC3 and PC4 connect to switch ports Fa0/3 and Fa0/4, these two ports belong to VLAN 20.
- The switch connects to the router through switch port Fa0/5, and this one port is configured as a trunk, carrying both VLAN 10 and VLAN 20 traffic together.
- The router's physical port Fa0/0 is split into two subinterfaces, Fa0/0.10 for VLAN 10 and Fa0/0.20 for VLAN 20.

The Full IP Address Plan

Device	VLAN	IP Address	Subnet Mask	Default Gateway
PC1	VLAN 10	192.168.1.10	255.255.255.0	192.168.1.1
PC2	VLAN 10	192.168.1.20	255.255.255.0	192.168.1.1
PC3	VLAN 20	192.168.2.10	255.255.255.0	192.168.2.1
PC4	VLAN 20	192.168.2.20	255.255.255.0	192.168.2.1
Router subinterface Fa0/0.10	VLAN 10 gateway	192.168.1.1	255.255.255.0	not applicable
Router subinterface Fa0/0.20	VLAN 20 gateway	192.168.2.1	255.255.255.0	not applicable

Notice that the router's subinterface address for each VLAN is exactly the same address used as the default gateway on that VLAN's PCs. That is not a coincidence, the router subinterface literally is the gateway.

Step 1, Assign IP Address and Default Gateway on Each PC

1. Click on PC1, go to Desktop, then IP Configuration.
2. Enter IP Address 192.168.1.10, Subnet Mask 255.255.255.0, and Default Gateway 192.168.1.1.
3. Repeat this for PC2, PC3, and PC4 using the exact values from the table above. This part does not use CLI at all, it is done through the graphical Desktop screen.

Step 2, Configure VLANs on the Switch, with Every Command Explained

Open the switch CLI as shown earlier. Type the following, one line at a time.

Create the two VLANs

Command	What It Actually Does
enable	Moves you from basic user mode into privileged mode, which allows deeper access to the device.
configure terminal	Enters global configuration mode, where device wide settings are changed.
vlan 10	Creates a new VLAN with the number 10, and enters VLAN configuration mode for it.
name SALES	Gives VLAN 10 a friendly, readable name, purely for humans, the switch does not need this to function.

vlan 20	Creates a second VLAN with the number 20.
name IT	Gives VLAN 20 a friendly name too.

Assign switch ports to the VLANs, as access ports

Command	What It Actually Does
interface fa0/1	Selects port Fa0/1 specifically, so the next commands only apply to this one port.
switchport mode access	Tells the switch that this port connects to a single normal device, not another switch or router.
switchport access vlan 10	Places this specific port inside VLAN 10.
interface fa0/2	Switches focus to port Fa0/2.
switchport mode access	Same meaning as above, applied to Fa0/2 now.
switchport access vlan 10	Places Fa0/2 inside VLAN 10 as well, this is PC2's port.
interface fa0/3	Switches focus to port Fa0/3.
switchport mode access	Marks Fa0/3 as an access port.
switchport access vlan 20	Places Fa0/3 inside VLAN 20, this is PC3's port.
interface fa0/4	Switches focus to port Fa0/4.
switchport mode access	Marks Fa0/4 as an access port.
switchport access vlan 20	Places Fa0/4 inside VLAN 20 too, this is PC4's port.

Shortcut you can use instead of repeating four separate blocks, interface range fa0/1-4 followed by switchport mode access, this selects all four ports at once and applies the access mode command to all of them together, then you would still assign VLANs individually or in smaller ranges.

Configure the trunk port toward the router

Command	What It Actually Does
interface fa0/5	Selects port Fa0/5, the one physically wired to the router.
switchport mode trunk	Marks this port as a trunk, allowing it to carry tagged traffic from multiple VLANs, VLAN 10 and VLAN 20 both, over this one cable.

Step 3, Configure Inter VLAN Routing on the Router, with Every Command Explained

Open the router CLI the same way, click the router, then the CLI tab.

Command	What It Actually Does
enable	Enters privileged mode on the router.
configure terminal	Enters global configuration mode on the router.
interface fa0/0	Selects the router's physical port that is wired to the switch's trunk port.
no shutdown	Physical router ports are turned off by default, this command turns the port on. Forgetting this single command is one of the most common reasons a whole lab fails, so never skip it.
interface fa0/0.10	Creates the first subinterface, numbered .10 to match VLAN 10, and enters its configuration mode. This does not need a separate no shutdown, since it rides on top of the already active Fa0/0.
encapsulation dot1q 10	Tells this subinterface to only handle traffic tagged as VLAN 10 by the trunk. The number 10 here must match the VLAN number exactly.
ip address 192.168.1.1	Gives this subinterface the IP address that VLAN 10 devices will use as their default gateway.

255.255.255.0	
interface fa0/0.20	Creates the second subinterface, numbered .20 to match VLAN 20.
encapsulation dot1q 20	Tells this subinterface to only handle traffic tagged as VLAN 20.
ip address 192.168.2.1 255.255.255.0	Gives this subinterface the IP address that VLAN 20 devices will use as their default gateway.

Notice the physical interface Fa0/0 itself never receives its own IP address in this setup, only its two subinterfaces do. This is intentional, the physical port is just the shared road, the subinterfaces are the actual doors into each VLAN.

Step 4, Save Your Configuration

Do this on both the switch and the router, so your work is not lost if the device restarts.

Command	What It Actually Does
end	Exits out of configuration mode, back to the privileged mode prompt.
copy running-config startup-config	Copies the currently active configuration into permanent storage, so it survives a restart. You may be asked to confirm a filename, just press Enter to accept the default.

Step 5, Testing, in the Correct Order

Test 1, same VLAN, should work immediately

Click PC1, go to Desktop, then Command Prompt, and type:

ping 192.168.1.20

This is pinging PC2, which is in the same VLAN as PC1. This should succeed even before the router is configured at all, because both devices are in the same broadcast domain and the switch alone can deliver traffic between them.

Test 2, different VLAN, before the router is configured

Still on PC1, type:

ping 192.168.2.10

This is pinging PC3, which is in a different VLAN. At this stage, before the router subinterfaces are set up, this ping is expected to fail completely, this is normal and expected, it proves that VLANs really do separate traffic.

Test 3, different VLAN, after the router is configured

Run the exact same command again:

ping 192.168.2.10

Now that the router has subinterfaces acting as gateways for both VLANs, this ping should succeed. The very first ping attempt may show Request timed out, this is because the devices are still learning each other's MAC addresses using ARP for the first time, simply run the ping again and it should show proper replies.

Troubleshooting, if a Ping Still Fails

- Check the cable between switch and router is plugged into Fa0/5 on the switch side and Fa0/0 on the router side, matching what the lab expects.
- On the router, confirm you typed no shutdown under interface fa0/0, this single missed step is the most common cause of total failure.
- Confirm the encapsulation dot1q number on each subinterface exactly matches the VLAN number, dot1q 10 for the .10 subinterface, dot1q 20 for the .20 subinterface.
- Confirm every PC has the correct default gateway matching its own VLAN, a PC in VLAN 10 must use 192.168.1.1, not 192.168.2.1.

- Confirm Fa0/5 on the switch is set to trunk mode, not access mode, otherwise it cannot carry both VLANs.
- On the switch, run **show vlan brief** to confirm each port landed in the VLAN you intended.
- On the router, run **show ip interface brief** to confirm both subinterfaces show an IP address and a status of up.
- Give it a second try, the very first ping after any configuration change commonly times out once due to ARP, and then works.

Viva Questions for This Practical

Q1. What is a VLAN?

A Virtual LAN, a way to logically separate devices into their own independent network using switch configuration, even if they share the same physical switch.

Q2. Why can devices in different VLANs not talk to each other by default?

Because each VLAN is its own separate broadcast domain and subnet, a switch alone cannot move traffic between two different subnets, only a router or a Layer 3 switch can.

Q3. What is the difference between an access port and a trunk port?

An access port belongs to exactly one VLAN and connects to a single end device. A trunk port can carry tagged traffic belonging to multiple VLANs at once, and is used to connect switches to each other or a switch to a router.

Q4. What does encapsulation dot1q do?

It tells a subinterface which VLAN's tagged traffic it should handle, based on matching the VLAN tag number carried inside each frame on the trunk.

Q5. What is Router on a Stick?

A design where a single physical router interface is divided into multiple subinterfaces, each one acting as the gateway for a different VLAN, allowing one physical cable to serve many VLANs.

Q6. Why does the router's physical interface Fa0/0 not get its own IP address in this setup?

Because the IP addresses are assigned to its subinterfaces instead, the physical interface only needs to be turned on using no shutdown, it acts purely as the shared physical path.

Q7. What is a default gateway and why does every PC need one here?

It is the address a device sends traffic to when the destination is outside its own subnet. Every PC needs one here so that traffic meant for another VLAN knows to go to the router first.

Q8. Why did the ping between VLANs fail before configuring the router, but succeed after?

Before router configuration, there was no device capable of moving traffic between the two separate subnets. After configuring the subinterfaces, the router became able to receive traffic from one VLAN and forward it into the other.

Q9. What command saves a device's configuration permanently?

copy running-config startup-config

Q10. What does no shutdown do and why is it important on Fa0/0 here?

It activates a router interface that is disabled by default. Without it, no traffic at all can pass through Fa0/0, even if the subinterfaces are perfectly configured, so this single line is critical.

Q11. What would happen if two PCs in different VLANs were accidentally given the same subnet, like both using 192.168.1.x?

It would cause addressing conflicts and broken routing, since each VLAN is meant to have its own unique subnet range for the router to correctly separate and forward traffic between them.

Q12. Which command shows which ports belong to which VLAN on the switch?

show vlan brief

Important Things to Remember

- Access ports connect to end devices and belong to one VLAN. Trunk ports connect to switches or routers and carry multiple VLANs at once.
- Every VLAN is its own subnet and its own broadcast domain, this is the entire reason a router is needed to connect them.
- The encapsulation dot1q number on a router subinterface must always match the VLAN number it is meant to serve.
- Never forget no shutdown on the router's physical interface, subinterfaces alone will not work if the parent physical interface is administratively down.

- Always save configuration on both the switch and the router using `copy running-config startup-config`, otherwise the setup is lost on restart.
- The very first ping after any fresh configuration change may time out due to ARP still learning addresses, this is normal, simply try again.
- Type commands one line at a time instead of pasting large blocks, it avoids CLI errors and helps you actually remember what each line does for the viva.

Quick recall order: PC IP and gateway, switch VLANs and ports, switch trunk port, router subinterfaces with no shutdown, save config, then test same VLAN, then test different VLAN.